

2 Unit-II: Limits and Continuity

Introduction

Consider the function

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad x \neq 2.$$

At $x = 2$, the expression is not defined because the denominator becomes zero. However, for $x \neq 2$,

$$x^2 - 4 = (x - 2)(x + 2),$$

and hence

$$f(x) = \frac{(x - 2)(x + 2)}{x - 2} = x + 2.$$

Now let us examine some values of $f(x)$ as x approaches 2.

x	$f(x)$
1.9	3.9
1.99	3.99
1.999	3.999
2.001	4.001
2.01	4.01
2.1	4.1

From the table, we observe that as x gets closer and closer to 2, the values of $f(x)$ get closer and closer to 4.

Therefore,

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = 4.$$

Even though the original function is not defined at $x = 2$, its limit exists and equals 4.

This shows that the limit depends on the behavior of the function near the point and not necessarily on the value of the function at the point.

The idea of a *limit* describes the value that a function approaches when the independent variable approaches a particular point.

Let f be a real-valued function defined in a neighborhood of a , except possibly at a itself. We say that

$$\lim_{x \rightarrow a} f(x) = L$$

if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a , with $x \neq a$.

Notice that the value $f(a)$ is not important for the existence of the limit. The limit concerns the behavior of $f(x)$ near a .

Definition, Meaning and Geometrical Interpretation

Definition 2.1 (ε - δ Definition). We say that

$$\lim_{x \rightarrow a} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$

Symbolically, $\forall \varepsilon > 0 \exists \delta > 0 \ni 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$

This is the rigorous definition of the limit of a function.

Meaning of ε :

The Greek letter

$$\varepsilon > 0$$

measures how close we want $f(x)$ to be to the proposed limit L .

The inequality

$$|f(x) - L| < \varepsilon$$

means that the distance between $f(x)$ and L is less than ε .

Equivalently,

$$L - \varepsilon < f(x) < L + \varepsilon.$$

Thus $f(x)$ lies inside the interval $(L - \varepsilon, L + \varepsilon)$.

This interval is called an ε -neighborhood of L .

Meaning of δ :

The number

$$\delta > 0$$

controls how close x must be to a .

The inequality

$$|x - a| < \delta$$

is equivalent to

$$a - \delta < x < a + \delta.$$

Because we exclude $x = a$, we write

$$0 < |x - a| < \delta.$$

Thus x lies in the deleted neighborhood

$$(a - \delta, a) \cup (a, a + \delta).$$

Geometrical Interpretation:

The definition

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$$

can be understood using two bands.

The condition

$$|x - a| < \delta$$

gives the vertical strip

$$a - \delta < x < a + \delta.$$

The condition

$$|f(x) - L| < \varepsilon$$

gives the horizontal strip

$$L - \varepsilon < y < L + \varepsilon.$$

Thus,

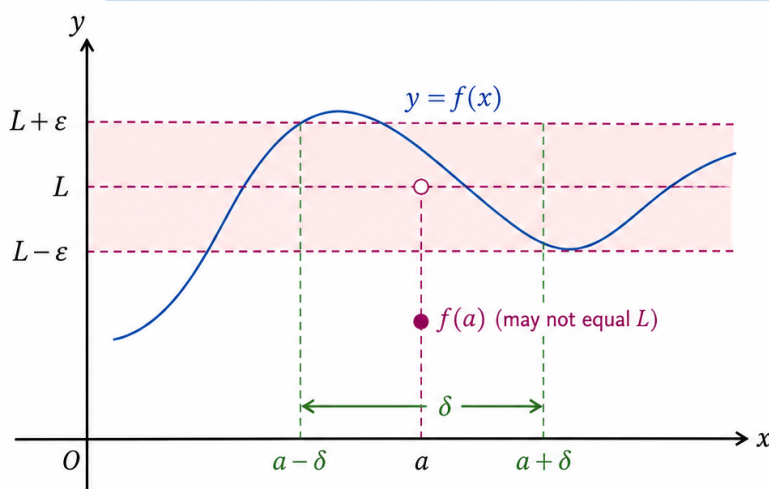
$$\lim_{x \rightarrow a} f(x) = L$$

means that, no matter how narrow a horizontal strip around L is chosen, we can find a sufficiently narrow vertical strip around a so that all points of the graph lying in that vertical strip, except possibly at $x = a$, lie inside the horizontal strip.

Geometrical Interpretation of a Limit

Let f be a function defined on an open interval containing a , except possibly at a .

We say $\lim_{x \rightarrow a} f(x) = L$ if and only if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a (but $x \neq a$).



Interpretation

- For any $\varepsilon > 0$, the horizontal band $L - \varepsilon < y < L + \varepsilon$ (the ε -neighbourhood of L) is shown by the shaded region.
- There exists $\delta > 0$ such that for all x with $0 < |x - a| < \delta$ (i.e., $a - \delta < x < a + \delta$, $x \neq a$), the points on the curve $y = f(x)$ lie inside the band $L - \varepsilon < y < L + \varepsilon$.
- The function value at $x = a$, $f(a)$, may be different from L (or $f(a)$ may not be defined).

Thus, $\lim_{x \rightarrow a} f(x) = L$ means that as x approaches a , the graph of $f(x)$ approaches the horizontal line $y = L$.

Why Do We Write $0 < |x - a|$?:

The condition

$$0 < |x - a|$$

means

$$x \neq a.$$

The value of the function at a itself does not determine the limit.

For instance,

$$f(x) = \begin{cases} x + 2, & x \neq 1, \\ 100, & x = 1. \end{cases}$$

Then

$$f(1) = 100,$$

but

$$\lim_{x \rightarrow 1} f(x) = 3.$$

Thus the limit depends on nearby values of the function, not necessarily on the value at the point.

Understanding the Logical Order:

The statement

$$\forall \varepsilon > 0, \quad \exists \delta > 0$$

is extremely important.

It means that first an arbitrary positive number ε is chosen. Then we must be able to find a suitable positive number δ .

Usually,

$$\delta = \delta(\varepsilon),$$

that is, δ may depend on ε , but δ must not depend on x .

The logical structure is

$$\boxed{\text{Given } \varepsilon > 0, \text{ find } \delta > 0 \text{ such that } |f(x) - L| < \varepsilon \text{ whenever } 0 < |x - a| < \delta.}$$

The same ε - δ definition may be stated as follows:

Definition 2.2 (Neighborhood Form of the Definition). For every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$f((a - \delta, a + \delta) \setminus \{a\}) \subseteq (L - \varepsilon, L + \varepsilon).$$

In words, every sufficiently small deleted neighborhood of a is mapped into the prescribed ε -neighborhood of L .

Examples Using ε - δ Definition

Problem 2.1. Prove that

$$\lim_{x \rightarrow 2} (3x + 1) = 7.$$

Solution

We want to show that for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x - 2| < \delta$$

implies

$$|(3x + 1) - 7| < \varepsilon.$$

Now,

$$|(3x + 1) - 7| = |3x - 6| = 3|x - 2|.$$

We want

$$3|x - 2| < \varepsilon.$$

This will happen if

$$|x - 2| < \frac{\varepsilon}{3}.$$

Therefore choose

$$\boxed{\delta = \frac{\varepsilon}{3}}.$$

Then, if

$$0 < |x - 2| < \delta,$$

we get

$$|(3x + 1) - 7| = 3|x - 2| < 3\delta = 3\left(\frac{\varepsilon}{3}\right) = \varepsilon.$$

Hence,

$$\boxed{\lim_{x \rightarrow 2} (3x + 1) = 7}.$$

Problem 2.2. Prove that

$$\lim_{x \rightarrow 3} x^2 = 9.$$

Solution

We have

$$|x^2 - 9| = |x - 3||x + 3|.$$

We want

$$|x - 3||x + 3| < \varepsilon.$$

Suppose we require

$$|x - 3| < 1.$$

Then

$$2 < x < 4.$$

Therefore,

$$5 < x + 3 < 7,$$

so

$$|x + 3| < 7.$$

Thus,

$$|x^2 - 9| = |x - 3||x + 3| < 7|x - 3|.$$

To make this less than ε , require

$$|x - 3| < \frac{\varepsilon}{7}.$$

Therefore choose

$$\delta = \min \left\{ 1, \frac{\varepsilon}{7} \right\}.$$

Then

$$0 < |x - 3| < \delta$$

implies

$$|x^2 - 9| < 7|x - 3| < 7\delta \leq \varepsilon.$$

Hence,

$$\lim_{x \rightarrow 3} x^2 = 9.$$

Problem 2.3. Prove that

$$\lim_{x \rightarrow 1} (x^2 + x) = 2.$$

Solution

We have

$$|x^2 + x - 2| = |(x - 1)(x + 2)| = |x - 1||x + 2|.$$

Suppose

$$|x - 1| < 1.$$

Then

$$0 < x < 2.$$

Hence,

$$2 < x + 2 < 4,$$

so

$$|x + 2| < 4.$$

Therefore,

$$|x^2 + x - 2| < 4|x - 1|.$$

To make this less than ε , require

$$|x - 1| < \frac{\varepsilon}{4}.$$

Hence choose

$$\delta = \min \left\{ 1, \frac{\varepsilon}{4} \right\}.$$

Then

$$0 < |x - 1| < \delta$$

implies

$$|x^2 + x - 2| < 4|x - 1| < 4\delta \leq \varepsilon.$$

Therefore,

$$\lim_{x \rightarrow 1} (x^2 + x) = 2.$$

Problem 2.4. Prove that

$$\lim_{x \rightarrow a} (mx + c) = ma + c.$$

Solution

Consider

$$f(x) = mx + c.$$

We want to prove that

$$\lim_{x \rightarrow a} f(x) = ma + c.$$

Let

$$L = ma + c.$$

Then

$$\begin{aligned} |f(x) - L| &= |(mx + c) - (ma + c)| \\ &= |m(x - a)| \\ &= |m||x - a|. \end{aligned}$$

To have

$$|f(x) - L| < \varepsilon,$$

it is sufficient that

$$|x - a| < \frac{\varepsilon}{|m|}.$$

Thus, for $m \neq 0$, choose

$$\delta = \frac{\varepsilon}{|m|}.$$

Therefore,

$$\lim_{x \rightarrow a} (mx + c) = ma + c.$$

Uniqueness of Limit

Theorem 2.1. *If $\lim_{x \rightarrow a} f(x)$ exists, then it is unique.*

Proof

Suppose that $f(x)$ has two limits L and M as $x \rightarrow a$.

Thus, $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} f(x) = M$.

We shall prove that $L = M$.

If possible, assume that $L \neq M$. Then $|L - M| > 0$.

Choose $\varepsilon = \frac{1}{2}|L - M| > 0$.

Since $\lim_{x \rightarrow a} f(x) = L$, by the definition of limit, there exists a number $\delta_1 > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta_1$.

Similarly, since $\lim_{x \rightarrow a} f(x) = M$, there exists a number $\delta_2 > 0$ such that $|f(x) - M| < \varepsilon$ whenever $0 < |x - a| < \delta_2$.

Let $\delta = \min\{\delta_1, \delta_2\}$.

Then, whenever $0 < |x - a| < \delta$, we have both $|f(x) - L| < \varepsilon$ and $|f(x) - M| < \varepsilon$.

Therefore, $|L - M| = |L - f(x) + f(x) - M| \leq |L - f(x)| + |f(x) - M|$ by the triangle inequality.

Hence, $|L - M| < \varepsilon + \varepsilon = 2\varepsilon$.

Using $\varepsilon = \frac{1}{2}|L - M|$, in the above, we obtain $|L - M| < 2 \cdot \frac{1}{2}|L - M| = |L - M|$, which is not possible.

Therefore, our assumption $L \neq M$ is false. Hence,

$$\boxed{L = M}.$$

Thus, if the limit of a function exists at a point, it is unique. □

Bounded Function

Definition 2.3.

1. **Bounded above:** The function f is said to be *bounded above* on A if there exists a real number M such that

$$f(x) \leq M \text{ for every } x \in A.$$

Such a number M is called an *upper bound* of f on A .

2. **Bounded below:** The function f is said to be *bounded below* on A if there exists a real number m such that

$$m \leq f(x) \text{ for every } x \in A.$$

Such a number m is called a *lower bound* of f on A .

3. **Bounded:** The function f is said to be *bounded* on A if it is both bounded above and bounded below. Thus, there exist real numbers m and M such that

$$m \leq f(x) \leq M \text{ for every } x \in A. \quad \text{(OR)}$$

Equivalently, there exists a real number $K > 0$ such that

$$|f(x)| \leq K, \text{ that is, } -K \leq f(x) \leq K \text{ for every } x \in A. \text{ (OR)}$$

Simply, a function is bounded if and only if its range is a bounded subset of $\mathbb{R}$.

Example 2.1. Consider $f(x) = \sin x$ on $\mathbb{R}$. Since $-1 \leq \sin x \leq 1$ for every $x \in \mathbb{R}$, we have $|\sin x| \leq 1$. Hence $f(x) = \sin x$ is bounded on $\mathbb{R}$.

Example 2.2. Consider $f(x) = \frac{1}{1+x^2}$ on $\mathbb{R}$. Since $x^2 \geq 0$, we have $1+x^2 \geq 1$. Therefore, $0 < \frac{1}{1+x^2} \leq 1$, which implies $-1 < 0 < \frac{1}{1+x^2} \leq 1$, Hence $\left| \frac{1}{1+x^2} \right| \leq 1$, and so f is bounded on $\mathbb{R}$.

Example 2.3. Consider $f(x) = x$ on the interval $[-2, 2]$. Then $-2 \leq x \leq 2$, and hence $|f(x)| = |x| \leq 2$. Therefore $f(x) = x$ is bounded on $[-2, 2]$. However, $f(x) = x$ is not bounded on $\mathbb{R}$, since its values can become arbitrarily large in both directions on $\mathbb{R}$.

Example 2.4. Consider $f(x) = \frac{1}{x}$ on the interval $(0, 1)$.

As x approaches 0, $\frac{1}{x}$ becomes arbitrarily large. Hence there is no $M > 0$ such that $\left| \frac{1}{x} \right| \leq M$ for every $x \in (0, 1)$.

Therefore, $f(x) = \frac{1}{x}$ is not bounded on $(0, 1)$.

Algebra of Limits

Lemma 2.1. If $\lim_{x \rightarrow a} f(x) = L$, then f is bounded in some deleted neighbourhood of a . Moreover, if $L \neq 0$, then $f(x) \neq 0$.

That is, there exist constants $M > 0$ and $\delta > 0$ such that $|f(x)| \leq M$ whenever $0 < |x - a| < \delta$ and if $L \neq 0$, then $f(x) \neq 0$.

Proof

Since $\lim_{x \rightarrow a} f(x) = L$, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$.

Now, by using the fact that $||c| - |d|| \leq |c - d|$, we have

$$||f(x)| - |L|| \leq |f(x) - L| < \varepsilon.$$

Therefore,

$$||f(x)| - |L|| < \varepsilon \quad \text{or} \quad -\varepsilon \leq |f(x)| - |L| < \varepsilon \quad \text{or} \quad |L| - \varepsilon \leq |f(x)| < |L| + \varepsilon. \quad (1)$$

Thus from right side inequality of (1), we have

$$|f(x)| < |L| + \varepsilon \implies f(x) \text{ is bounded, whenever } 0 < |x - a| < \delta.$$

Further if $L \neq 0$, take $\varepsilon = \frac{|L|}{2}$, therefore from (1), we have

$$|L| - \frac{|L|}{2} < |f(x)| < |L| + \frac{|L|}{2} \quad \text{or} \quad \frac{|L|}{2} < |f(x)| < \frac{3|L|}{2}, \text{ which implies that } f(x) \neq 0.$$

□

Theorem 2.2. *If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then $\lim_{x \rightarrow a} (f(x) + g(x)) = L + M$.*
(KUD: Jan-2026, Feb-2025, Feb-2024, Mar-2022)(Question 1-QB)

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_1$.

Since $\lim_{x \rightarrow a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_2$.

Choose $\delta = \min\{\delta_1, \delta_2\}$.

Then, whenever $0 < |x - a| < \delta$, we have

$$\begin{aligned} |(f(x) + g(x)) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M| \quad (\because |c + d| \leq |c| + |d|) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Hence,

$$\boxed{\lim_{x \rightarrow a} (f(x) + g(x)) = L + M}.$$

□

Theorem 2.3. *If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then $\lim_{x \rightarrow a} (f(x) - g(x)) = L - M$.*
(KUD: Feb-2023, Oct-2022)(Question 2-QB)

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_1$.

Since $\lim_{x \rightarrow a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_2$.

Choose $\delta = \min\{\delta_1, \delta_2\}$.

Then, whenever $0 < |x - a| < \delta$, we have

$$\begin{aligned} |(f(x) - g(x)) - (L - M)| &= |(f(x) - L) + (-g(x) + M)| \\ &\leq |f(x) - L| + |-g(x) + M| \quad (\because |c + d| \leq |c| + |d|) \\ &\leq |f(x) - L| + |g(x) - M| \quad (\because |-c| = |c|) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Therefore,

$$\boxed{\lim_{x \rightarrow a} (f(x) - g(x)) = L - M}.$$

□

Theorem 2.4. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then $\lim_{x \rightarrow a} f(x)g(x) = LM$.

(Question 3-QB)

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon}{2(|M| + 1)}$ whenever $0 < |x - a| < \delta_1$.

Since $\lim_{x \rightarrow a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon}{2(|L| + 1)}$ whenever $0 < |x - a| < \delta_2$.

Also, since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_3 > 0$ such that $|f(x)| < |L| + 1$ whenever $0 < |x - a| < \delta_3$. (see, Lemma 1 for $\varepsilon = 1$)

Choose $\delta = \min\{\delta_1, \delta_2, \delta_3\}$.

Then, whenever $0 < |x - a| < \delta$, we have

$$\begin{aligned}
 |f(x)g(x) - LM| &= |f(x)g(x) - f(x)M + f(x)M - LM| \\
 &= |f(x)(g(x) - M) + M(f(x) - L)| \\
 &\leq |f(x)| |g(x) - M| + |M| |f(x) - L| \\
 &< (|L| + 1) \frac{\varepsilon}{2(|L| + 1)} + |M| \frac{\varepsilon}{2(|M| + 1)} \\
 &< \frac{\varepsilon}{2} + 1 \cdot \frac{\varepsilon}{2} \\
 &= \varepsilon.
 \end{aligned}$$

Hence,

$$\boxed{\lim_{x \rightarrow a} f(x)g(x) = LM}.$$

□

Theorem 2.5. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, where $M \neq 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon|M|}{4}$ whenever $0 < |x - a| < \delta_1$.

Since $\lim_{x \rightarrow a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon|M|^2}{4(|L| + 1)}$ whenever $0 < |x - a| < \delta_2$.

Also, since $\lim_{x \rightarrow a} g(x) = M$ and $M \neq 0$, there exists $\delta_3 > 0$ such that $\frac{|M|}{2} < |g(x)|$ whenever $0 < |x - a| < \delta_3$. (see, Lemma 1)

Choose $\delta = \min\{\delta_1, \delta_2, \delta_3\}$.

Then, whenever $0 < |x - a| < \delta$, we have

$$\begin{aligned}
 \left| \frac{f(x)}{g(x)} - \frac{L}{M} \right| &= \left| \frac{Mf(x) - Lg(x)}{Mg(x)} \right| = \left| \frac{Mf(x) - LM + LM - Lg(x)}{Mg(x)} \right| \\
 &= \left| \frac{M(f(x) - L) + L(M - g(x))}{Mg(x)} \right| \\
 &\leq \frac{|M| |f(x) - L| + |L| |g(x) - M|}{|M| |g(x)|}.
 \end{aligned}$$

Since $|g(x)| > \frac{|M|}{2} \implies \frac{1}{|g(x)|} < \frac{2}{|M|}$, we obtain

$$\begin{aligned} \left| \frac{f(x)}{g(x)} - \frac{L}{M} \right| &< \frac{2}{|M|^2} (|M| |f(x) - L| + |L| |g(x) - M|) \\ &< \frac{2}{|M|^2} \left(|M| \frac{\varepsilon |M|}{4} + |L| \frac{\varepsilon |M|^2}{4(|L| + 1)} \right) \\ &= \frac{\varepsilon}{2} + \frac{|L|}{(|L| + 1)} \frac{\varepsilon}{2} < \frac{\varepsilon}{2} + 1 \cdot \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Therefore,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

□

Theorem 2.6. If $\lim_{x \rightarrow a} f(x) = L$, then $\lim_{x \rightarrow a} |f(x)| = |L|$.

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$.

Therefore, whenever $0 < |x - a| < \delta$, we have $||f(x)| - |L|| \leq |f(x) - L| < \varepsilon$.

Hence,

$$\lim_{x \rightarrow a} |f(x)| = |L|.$$

□

Note (Converse:). The converse of the above theorem is *not true*.

That is, the statement $\lim_{x \rightarrow a} |f(x)| = |L|$ does not necessarily imply that $\lim_{x \rightarrow a} f(x) = L$.

Example 2.5 (Counterexample).

Define $f(x) = \begin{cases} 1, & x > 0, \\ -1, & x < 0. \end{cases}$

We have $|f(x)| = 1$ for every $x \neq 0$. Therefore, $\lim_{x \rightarrow 0} |f(x)| = 1$.

However, $\lim_{x \rightarrow 0^+} f(x) = 1$ and $\lim_{x \rightarrow 0^-} f(x) = -1$.

Since, the right-hand and left-hand limits are different, $\lim_{x \rightarrow 0} f(x)$ does not exist.

Hence, the converse of the above theorem is false.

$\therefore \boxed{\lim_{x \rightarrow a} f(x) = L \implies \lim_{x \rightarrow a} |f(x)| = |L|}$, but $\boxed{\lim_{x \rightarrow a} |f(x)| = |L| \not\implies \lim_{x \rightarrow a} f(x) = L}$.

Continuity

Intuitively, a function is continuous at a point if small changes in the input near that point produce small changes in the output. Thus, the values of $f(x)$ can be made arbitrarily close to $f(a)$ by taking x sufficiently close to a .

At a point a where the limit is applicable, continuity means

$$\boxed{\lim_{x \rightarrow a} f(x) = f(a).} \quad \text{That is,} \quad \boxed{\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a).}$$

In particular, $f(a)$ must be defined, the limit must exist, and the limit must equal $f(a)$.

Definition 2.4 (ε - δ Definition of Continuity). Let $f : D \rightarrow \mathbb{R}$ and let $a \in D$. We say that f is **continuous at** a if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - f(a)| < \varepsilon$ whenever $x \in D$ and $|x - a| < \delta$.

Symbolically, $\boxed{\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D, |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.}$

(Question 1-QB)

Remark. The conditions $|x - a| < \delta$ and $0 < |x - a| < \delta$ give equivalent definitions of continuity at a , provided that $f(a)$ is defined.

Indeed, if $|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$, then the same implication clearly holds when $0 < |x - a| < \delta$.

Conversely, suppose that $0 < |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$. For $|x - a| < \delta$, there are two cases:

- If $x \neq a$, then $0 < |x - a| < \delta$, so $|f(x) - f(a)| < \varepsilon$ by assumption.
- If $x = a$, then $|f(x) - f(a)| = |f(a) - f(a)| = 0 < \varepsilon$.

Hence the inequality holds whenever $|x - a| < \delta$. Thus, including or excluding $x = a$ does not change the definition of continuity.

A function is **continuous on** D if it is continuous at every point of D .

Continuity on a closed interval: A function $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ if it is continuous at every point of (a, b) , right-continuous at a , and left-continuous at b . Thus, at the endpoints, we require $\lim_{x \rightarrow a^+} f(x) = f(a)$ and $\lim_{x \rightarrow b^-} f(x) = f(b)$.

Theorem 2.7 (Subdivision Theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then, for every $\varepsilon > 0$, there exists a finite partition $a = a_0 < a_1 < \dots < a_m = b$ such that $|f(u) - f(v)| < \varepsilon$ whenever u and v belong to the same subinterval $[a_{j-1}, a_j]$.

Proof

Fix $\varepsilon > 0$. Call a finite partition *suitable* if $|f(u) - f(v)| < \varepsilon$ whenever u and v belong to the same subinterval.

Suppose, for a contradiction, that $[a, b]$ has no suitable finite partition. Bisect $[a, b]$ into $\left[a, \frac{a+b}{2}\right]$ and $\left[\frac{a+b}{2}, b\right]$. If both halves had suitable finite partitions, combining

their partition points would give a suitable finite partition of $[a, b]$. Therefore, at least one half has no suitable finite partition. Choose such a half and denote it by I_1 . Then $|I_1| = \frac{b-a}{2}$, where $|I_1|$ denotes the length of I_1 .

Now bisect I_1 . By the same reasoning, at least one of its halves has no suitable finite partition. Choose such a half and denote it by I_2 . Continuing this process, we obtain nested closed intervals

$$[a, b] \supseteq I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \supseteq \cdots$$

such that each I_n has no suitable finite partition and $|I_n| = \frac{b-a}{2^n}$.

By the nested interval property, there exists a point c belonging to every I_n . In particular, $c \in [a, b]$. Since f is continuous at c , there exists $\delta > 0$ such that

$$x \in [a, b] \text{ and } |x - c| < \delta \implies |f(x) - f(c)| < \frac{\varepsilon}{2}.$$

Choose n sufficiently large that

$$|I_n| = \frac{b-a}{2^n} < \delta.$$

Since c and any point $x \in I_n$ lie in the same interval, their distance cannot exceed its length. Thus

$$|x - c| \leq |I_n| < \delta.$$

Consequently,

$$|f(x) - f(c)| < \frac{\varepsilon}{2} \quad \text{for every } x \in I_n.$$

For any $u, v \in I_n$, the triangle inequality now gives

$$\begin{aligned} |f(u) - f(v)| &\leq |f(u) - f(c)| + |f(c) - f(v)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

This contradicts the choice of I_n as an interval having no suitable finite partition. Hence $[a, b]$ has the required finite partition. $\square$

Theorem 2.8. *Every continuous function defined on $[a, b]$ is bounded on $[a, b]$.*

(KUD: Feb-2025)(Question 6-QB)

Proof

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous.

Let $\varepsilon > 0$ be given. Then there exists a finite partition $a = a_0 < a_1 < \cdots < a_n = b$ such that $|f(u) - f(v)| < \varepsilon$ whenever u and v belong to the same subinterval.

Let $x \in [a, b]$. Then $x \in [a_{k-1}, a_k]$ for some $k \in \{1, \dots, n\}$. We can write

$$\begin{aligned} f(x) &= f(x) + f(a_0) \underbrace{- f(a_0) + f(a_1)}_{\text{}} \underbrace{- f(a_1) + f(a_2)}_{\text{}} \underbrace{- f(a_2) + \dots}_{\text{}} \\ &\quad \underbrace{\dots + f(a_{k-1})}_{\text{}} \underbrace{- f(a_{k-1}) + f(a_k)}_{\text{}} - f(a_k) \\ &= f(a_0) + \sum_{j=1}^k (f(a_j) - f(a_{j-1})) + (f(x) - f(a_k)). \end{aligned}$$

By the triangle inequality, we have

$$\begin{aligned} |f(x)| &\leq |f(a_0)| + \sum_{j=1}^k |f(a_j) - f(a_{j-1})| + |f(x) - f(a_k)| \\ &< |f(a_0)| + k\varepsilon + \varepsilon \\ &= |f(a)| + (k+1)\varepsilon \\ &\leq |f(a)| + (n+1)\varepsilon, \text{ which is independent of } x. \end{aligned}$$

Thus, setting $M = |f(a)| + (n+1)\varepsilon$, we obtain $|f(x)| \leq M$ for every $x \in [a, b]$. Therefore, f is bounded on $[a, b]$. $\square$

Definition 2.5 (Supremum and Infimum of a Function). Let f be a real-valued function defined on a nonempty set A .

1. **Supremum.** If f is bounded above on A , the **supremum** of f on A is the least upper bound of its range, denoted by $\sup_{x \in A} f(x)$ or, simply $\sup f(x)$.

Thus, $M = \sup_{x \in A} f(x)$ if:

- (a) $f(x) \leq M$ for every $x \in A$;
- (b) for every $\varepsilon > 0$, there exists $x \in A$ such that $M - \varepsilon < f(x) \leq M$.

✚ : If $f(c) = M$ for some $c \in A$, the supremum is **attained**, and M is called the **maximum value** of f on A .

2. **Infimum.** If f is bounded below on A , the **infimum** of f on A is the greatest lower bound of its range, denoted by $\inf_{x \in A} f(x)$. or, simply $\inf f(x)$.

Thus, $m = \inf_{x \in A} f(x)$ if:

- (a) $m \leq f(x)$ for every $x \in A$;
- (b) for every $\varepsilon > 0$, there exists $x \in A$ such that $m \leq f(x) < m + \varepsilon$.

✚ : If $f(d) = m$ for some $d \in A$, the infimum is **attained**, and m is called the **minimum value** of f on A .

Example 2.6 (Supremum and Infimum).

1. **Both supremum and infimum are attained.**

Let $f(x) = x^2$ on $[0, 2]$. Its range is $[0, 4]$. Therefore,

$$\sup_{x \in [0, 2]} f(x) = 4 \quad \text{and} \quad \inf_{x \in [0, 2]} f(x) = 0.$$

Since $f(2) = 4$ and $f(0) = 0$, the supremum is attained at $x = 2$, and the infimum is attained at $x = 0$. Thus, the maximum and minimum values of $f(x)$ on $[0, 2]$ are 4 and 0, respectively.

2. **Neither supremum nor infimum is attained.**

Let $f(x) = x^2$ on $(0, 2)$. Its range is $(0, 4)$. Therefore,

$$\sup_{x \in (0, 2)} f(x) = 4 \quad \text{and} \quad \inf_{x \in (0, 2)} f(x) = 0.$$

The values of $f(x)$ approach 4 as $x \rightarrow 2^-$ and 0 as $x \rightarrow 0^+$, but

$$0 < f(x) < 4 \quad \text{for every } x \in (0, 2).$$

Hence, neither bound is attained, and f has neither a maximum nor a minimum value on $(0, 2)$.

3. **Supremum is attained, but infimum is not attained.**

Let $f(x) = 1 - x^2$ on $(-1, 1)$. Its range is $(0, 1]$. Therefore,

$$\sup_{x \in (-1, 1)} f(x) = 1 \quad \text{and} \quad \inf_{x \in (-1, 1)} f(x) = 0.$$

Since $f(0) = 1$, the supremum is attained at $x = 0$. However, $f(x) > 0$ for every $x \in (-1, 1)$, and $f(x) \rightarrow 0$ as $x \rightarrow 1^-$. Thus, the maximum value is 1, but there is no minimum value.

4. **Infimum is attained, but supremum is not attained.**

Let $f(x) = \frac{x}{1+x}$ on $[0, \infty)$. Its range is $[0, 1)$. Therefore,

$$\sup_{x \in [0, \infty)} f(x) = 1 \quad \text{and} \quad \inf_{x \in [0, \infty)} f(x) = 0.$$

Since $f(0) = 0$, the infimum is attained at $x = 0$. However, $f(x) < 1$ for every $x \geq 0$, and $f(x) \rightarrow 1$ as $x \rightarrow \infty$. Thus, the minimum value is 0, but there is no maximum value of $f(x)$ on $[0, \infty)$.

Remark. The supremum and infimum need not belong to the range of f . When attained, they are the maximum and minimum values of f on given domain of f , respectively.

Theorem 2.9 (Extreme Value Theorem). *Prove that a function continuous on $[a, b]$ attains its bounds.* (OR)

Prove that a function continuous on $[a, b]$ attains its supremum and infimum at least once in $[a, b]$.

(KUD: Jan-2026)(Question 8-QB)

Proof

Since f is continuous on $[a, b]$, the boundedness theorem implies that f is bounded on $[a, b]$. Hence its supremum and infimum exist as real numbers. Let on $[a, b]$

$$M = \sup f(x) \quad \text{and} \quad m = \inf f(x).$$

Suppose, for a contradiction, that f does not attain M .

Since $f(x) \leq M$, we then have $f(x) < M$ for every $x \in [a, b]$.

Thus $M - f(x) > 0$ for every $x \in [a, b]$, so the function $g(x) = \frac{1}{M - f(x)}$ is continuous on $[a, b]$.

By the boundedness theorem, g is bounded. Therefore, there exists a real number $K > 0$ such that $\frac{1}{M - f(x)} \leq K$ for every $x \in [a, b]$.

It follows that $M - f(x) \geq \frac{1}{K}$, and hence $f(x) \leq M - \frac{1}{K} < M$ for every $x \in [a, b]$.

Thus $M - \frac{1}{K}$ is an upper bound for f on $[a, b]$ smaller than M , contradicting the fact that M is the least upper bound.

Therefore, there exists $\alpha \in [a, b]$ such that $f(\alpha) = M$.

$\implies f$ attains supremum at least once on $[a, b]$.

Similarly, one can prove that f attains infimum at least once on $[a, b]$. $\square$

Theorem 2.10 (Intermediate Value Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, with $f(a) \neq f(b)$. Then f takes every value between $f(a)$ and $f(b)$ at least once in (a, b) . (OR)*

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, with $f(a) \neq f(b)$. Then for every real number k between $f(a)$ and $f(b)$, there exists $c \in (a, b)$ such that $f(c) = k$.

(KUD: Feb-2025)(Question 7-QB)

Proof

Suppose first that $f(a) < f(b)$, and let k be any number such that $f(a) < k < f(b)$. Define $g(x) = f(x) - k$ on $[a, b]$.

Since f is continuous and k is a constant, g is continuous on $[a, b]$.

Moreover, $g(a) = f(a) - k < 0$ and $g(b) = f(b) - k > 0$.

Thus $g(a)$ and $g(b)$ have opposite signs.

$\implies$ there exists $c \in (a, b)$ such that $g(c) = 0$.

Therefore, $g(c) = f(c) - k = 0 \implies f(c) = k$.

Again, if $f(a) > f(b)$, take k such that $f(b) < k < f(a)$. Then $g(a) > 0$ and $g(b) < 0$, so again there exists $c \in (a, b)$ with $f(c) = k$.

Hence f takes every value between $f(a)$ and $f(b)$ at least once in (a, b) . $\square$

Problem 2.5. Show that $f(x) = x^2$ takes the value 5 on $[1, 3]$ using the Intermediate Value Theorem.

(Question 9(i)-QB)

Solution

The function $f(x) = x^2$ is a polynomial, so it is continuous on $[1, 3]$.

Also, $f(1) = (1)^2 = 1$ and $f(3) = (3)^2 = 9$.

Since $f(1) = 1 < 5 < 9 = f(3)$, the Intermediate Value Theorem guarantees that there exists $c \in (1, 3)$ such that $f(c) = 5$.

To find c , we solve $f(c) = c^2 = 5 \implies c = \pm\sqrt{5}$.

Since $c \in (1, 3)$, we obtain $\boxed{c = \sqrt{5}}$.

Thus, $f(x) = x^2$ takes the value 5 at $x = \sqrt{5} \in [1, 3]$.

Problem 2.6. Show that $f(x) = \sin x$ takes the value 0.5 on $\left[0, \frac{\pi}{2}\right]$ using the Intermediate Value Theorem.

(Question 9(iv)-QB)

Solution

The function $f(x) = \sin x$ is continuous on $\left[0, \frac{\pi}{2}\right]$.

Also, $f(0) = \sin 0 = 0$ and $f\left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} = 1$.

Since $f(0) = 0 < 0.5 < 1 = f\left(\frac{\pi}{2}\right)$, the Intermediate Value Theorem guarantees that there exists $c \in \left(0, \frac{\pi}{2}\right)$ such that $f(c) = 0.5$.

To find c , we solve $f(c) = \sin c = \frac{1}{2} \implies \boxed{c = \frac{\pi}{6}}$.

Thus, $f(x) = \sin x$ takes the value 0.5 at $x = \frac{\pi}{6} \in \left[0, \frac{\pi}{2}\right]$.

Problem 2.7. Show that $x^2 - 2x - 5 = 0$ has at least one root in $(3, 4)$ using the Intermediate Value Theorem.

(Question 10(i)-QB)

Solution

Let $f(x) = x^2 - 2x - 5$. Since f is a polynomial, it is continuous on $[3, 4]$.

Also, $f(3) = 9 - 6 - 5 = -2$ and $f(4) = 16 - 8 - 5 = 3$.

Since $f(3) = -2 < 0 < 3 = f(4)$, the Intermediate Value Theorem guarantees that there exists $c \in (3, 4)$ such that $f(c) = 0$. Hence, $x^2 - 2x - 5 = 0$ has at least one root in $(3, 4)$.

To find c , we solve $c^2 - 2c - 5 = 0$ using the quadratic formula. Therefore,

$$\begin{aligned} c &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-5)}}{2(1)} = \frac{2 \pm \sqrt{4 + 20}}{2} \\ &= \frac{2 \pm \sqrt{24}}{2} = \frac{2 \pm 2\sqrt{4 \times 6}}{2} = \frac{2 \pm 2\sqrt{6}}{2} = 1 \pm \sqrt{6} \end{aligned}$$

Since $c \in (3, 4)$, we obtain $\boxed{c = 1 + \sqrt{6}}$.

Thus, $x^2 - 2x - 5 = 0$ has a root $x = 1 + \sqrt{6} \in (3, 4)$.